# Plan for Today

- Working with ERT data (Kleinsee)

  *from acquisition – to processing – to modelling – to visualization*

# From Data Acquisition to Visualization of Subsurface Resistivity Models

**Workflow of the processes involved in handling ERT geophysical data**

## Acquisition → Processing → Modelling → Visualization

- Electre: Configures and acquires the data in the field. (IRIS Syscal Pro)
- Prosys II: Filters and preprocesses the acquired data.
- Res2DInv: Performs the inversion of the preprocessed data to obtain a subsurface resistivity models
- Surfer: Visualizes the subsurface resistivity model generated

# Data Acquisition → Multi-electrode geoelectrical apparatus IRIS Syscal Pro Switch


**Syscal**

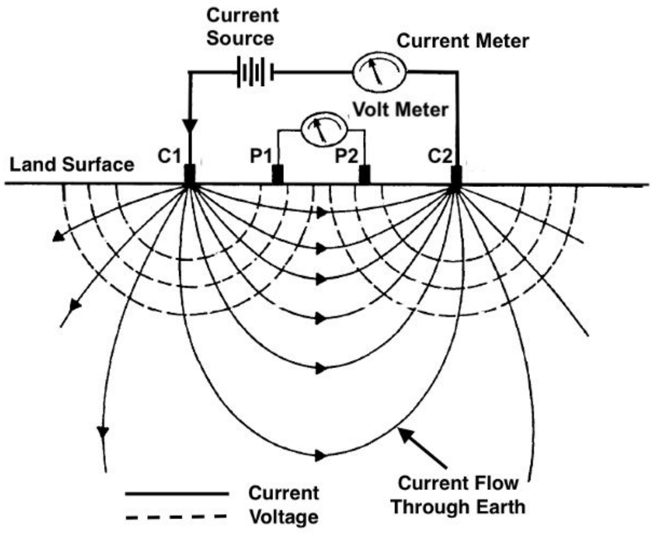
- Supports up to 48 electrodes (e.g. 10 m spacing), ideal for large-scale ERT/IP surveys
- High-power transmitter: 250 W, 2000 V peak-to-peak
- 10-channel receiver: Enables simultaneous data acquisition → faster and more efficient

**Electre**

We can control acquisition parameters, also manages sequences.

- Array configuration (Wenner, Schlumberger, Dipole–Dipole)
- Measurement sequence: injection time, current level
- Data management:
  - Real-time monitoring
  - Stores raw data in .bin format

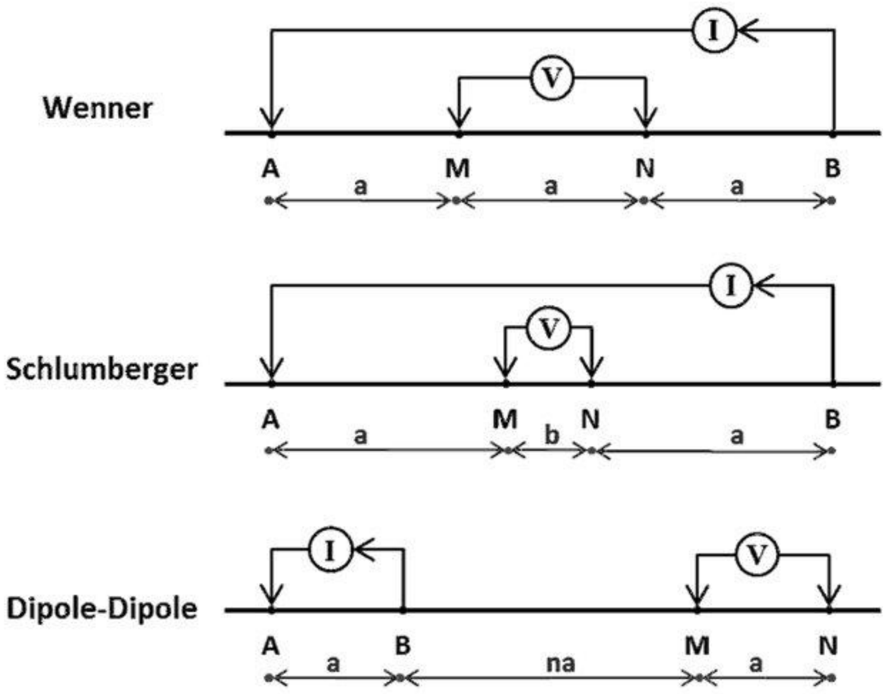
# Remembering how the method works




- Uses 4 electrodes, C1 and C2 (current electrodes) and P1 and P2 (potential electrodes)
- We measure Voltage V, and Current I
- Method is based on *Ohm's Law* $$\rho_a = k\frac{V}{I}$$
- Under consideration of the geometric position of the electrodes (K-factor) we obtain the resistivity

  $\rho$a: apparent resistivity (Ωm)
  k: Geometric factor, represents the distance between the different electrodes
  U: Potential (V)
  I: Current (A)
- Then we move the electrodes and we repeat the procedure

# Electrode Array




**Wenner configuration**

- Equal spacing between all electrodes (A-M-N-B)
- String signal - Good vertical resolution
- Less sensitive to lateral changes → Useful for detecting horizontal layers

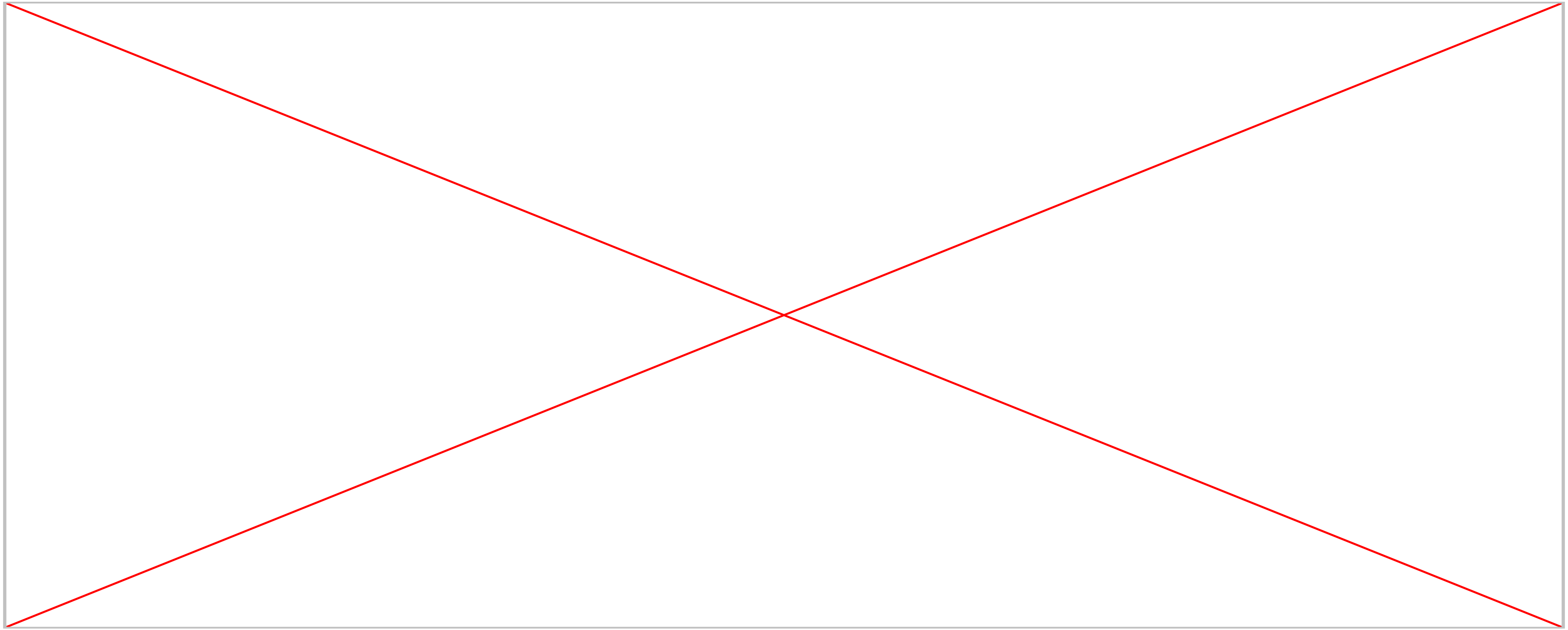
**Schlumberger configuration**

- Potential electrodes (M–N) are close together; current electrodes (A–B) are expanded symmetrically.
- Greater depth of investigation than Wenner
- Lower lateral resolution than dipole dipole
- Good vertical resolution: sensitive to resistivity changes with depth.

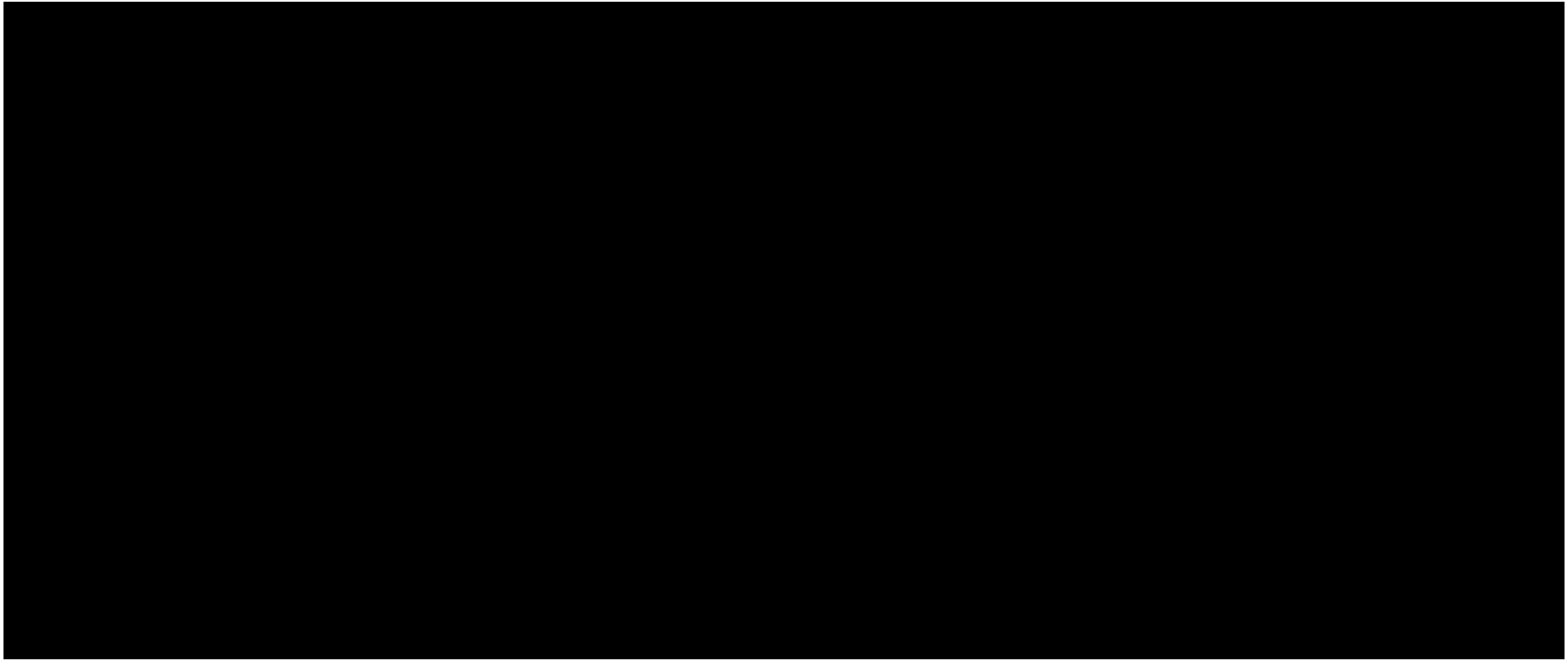
**Dipole Dipole configuration**

- Both current (A-B) and potential electrodes(M-N) are in tight pairs, moved progressively along the line
- High sensitivity to lateral variations, but signal is waker

# Example: Wenner configuration



# Example: Dipole-Dipole configuration



# Electrode Array Effect – Depth & Resolution

What are we looking for? How deep do you we to go?

That determines

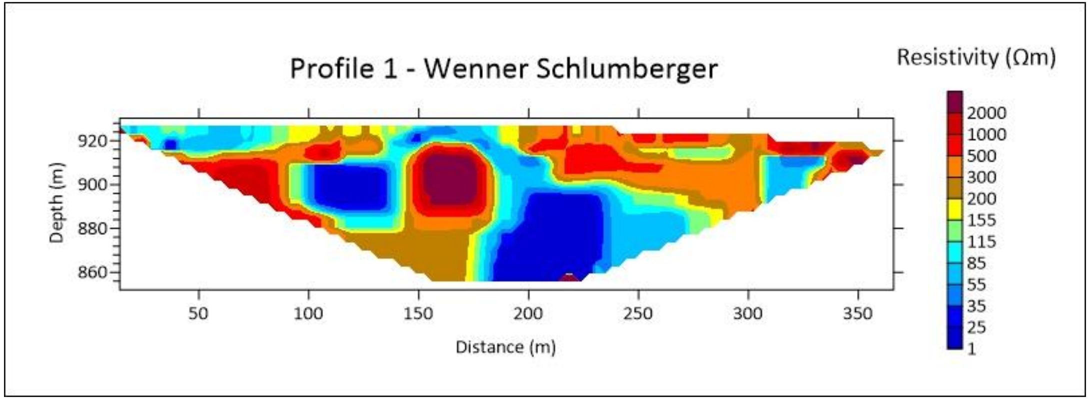
- The electrode array type
- The spacing between electrodes

Larger spacing ⟶ greater depth but lower resolution

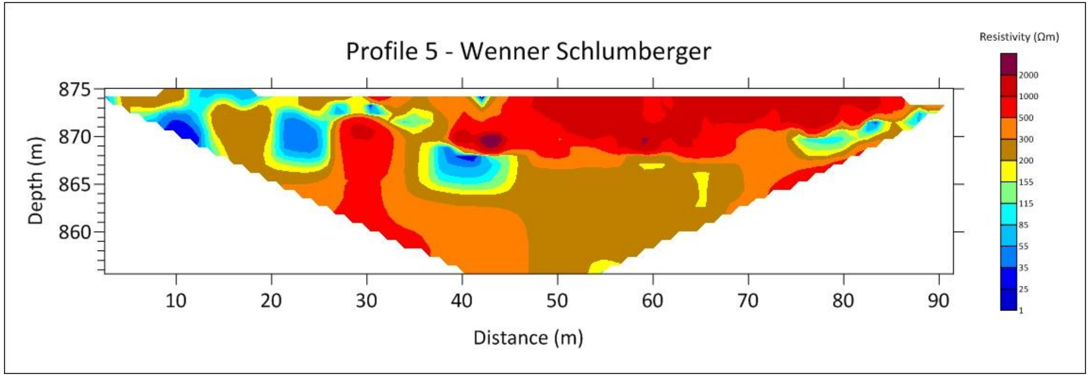
Smaller spacing ⟶ shallower depth but higher resolution

Trade-off between depth of investigation and ability to resolve small features

# Examples from Petra: Effect of Electrode Spacing on Depth

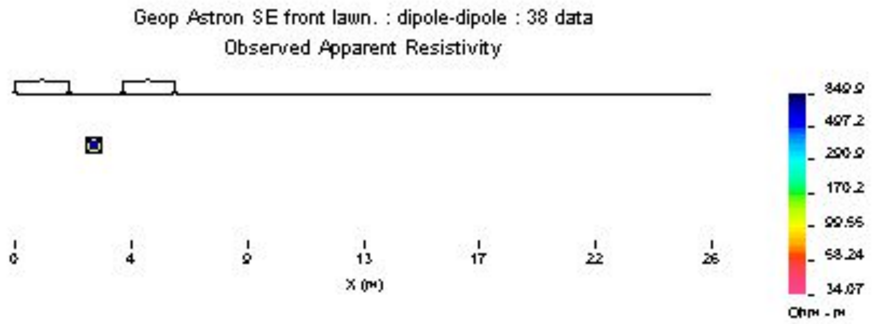



Profile 1 : 8 m electrode spacing




Profile 5: 2 m electrode spacing

# How do we estimate the resistivity? ⟶ Apparent Resistivity



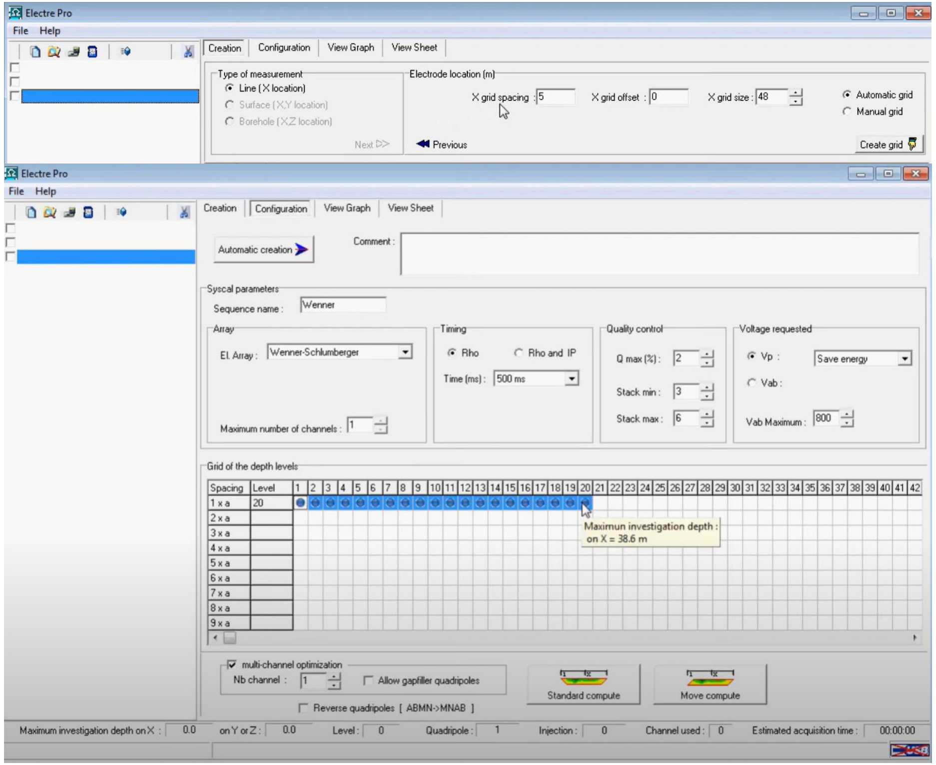

Example on the whiteboard

$$\rho_a = k\frac{V}{I}$$

ρa: apparent resistivity (Ωm)
k: Geometric factor, represents the distance between the different electrodes
U: Potential (V)
I: Current (A)

It is based on the assumption that the ground is homogeneous and isotropic (homogeneous half space)

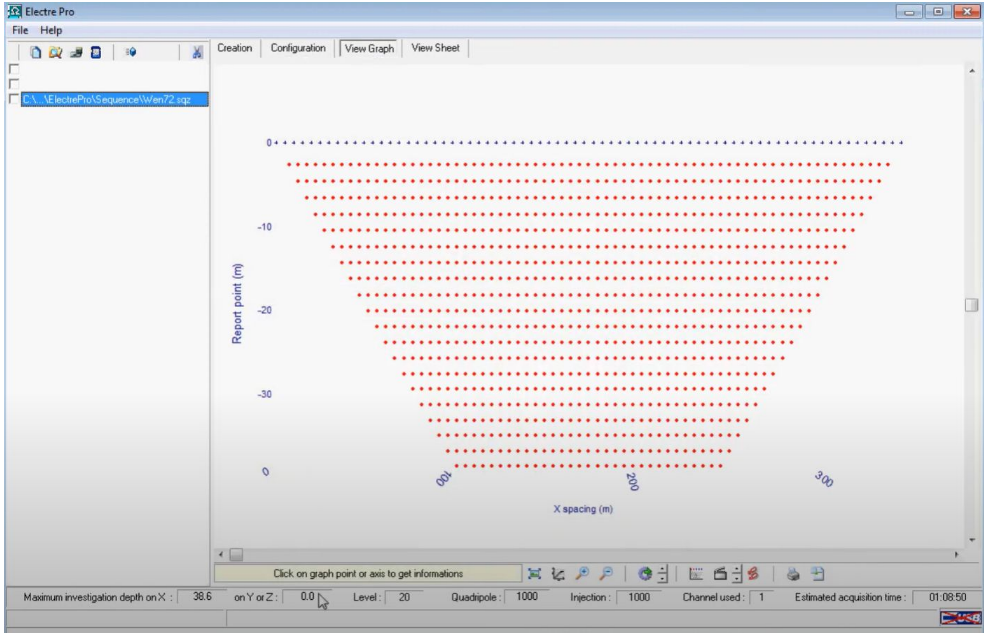
# What can I configure with a sequence file?




**Main parameters I can set**

- Electrode array (e.g., Wenner, Schlumberger, Dipole–Dipole)
- Electrode spacing (defines resolution and investigation depth)
- Measurement type / geometry (2D line, surface grid, borehole)
- Target depth and resolution goal
- Voltage range limits (Vρ and Vab for quality control)
- Stacking factor (number of repeated measurements per point)
- Acquisition time per point
- Energy mode (standard vs. energy-saving)

# Previewing the Sequence → Depth and Data Coverage




- **Penetration depth**:
  The deepest points define the **maximum investigation depth**, depending on electrode spacing and selected levels.
- **Data density**:
  The **spacing between horizontal lines** shows the **vertical resolution**:
  - Tighter spacing = higher resolution
  - Wider spacing = lower resolution at depth
- **Lateral coverage**:
  The width of the spread shows how much of the profile will be covered by measurements.

# From Data Acquisition to Visualization of Subsurface Resistivity Models

**Workflow of the processes involved in handling ERT geophysical data**

Acquisition → Processing → Modelling → Visualization

- Electre: Configures and acquires the data in the field.
- Prosys II: Filters and preprocesses the acquired data. → We want to improve data quality!
- Res2DInv: Performs the inversion of the preprocessed data to obtain a subsurface resistivity models
- Surfer: Visualizes the subsurface resistivity model generated

# Processing Data → Prosys II

**Values of interest**

- Rho: resistivity value (in Ohm.m)
- Dev: standard deviation (quality factor, in %)

- Multiple apparent resistivity values were measured at the same point → this leads to standard deviation
- If the standard deviation is low, this indicates consistent measurements.
- If the standard deviation is high, this suggests greater variability which is possibly due to noise in the data or issues during acquisition.

**Elimination of Bad Data**

- Use *Processing* → *Exterminate bad data points* to visualize and remove outliers.
- Automatic filtering based on standard deviation (e.g., 20% threshold).
- Visual check using Rho vs. depth plots.

**Smoothing Techniques (Lateral Filtering)**

**Sliding Average**

- Lateral filter (along X).
- Applied per depth level (not in Z).
- Uses a weighted average within a defined window.
- Smooths lateral variation without mixing depths.

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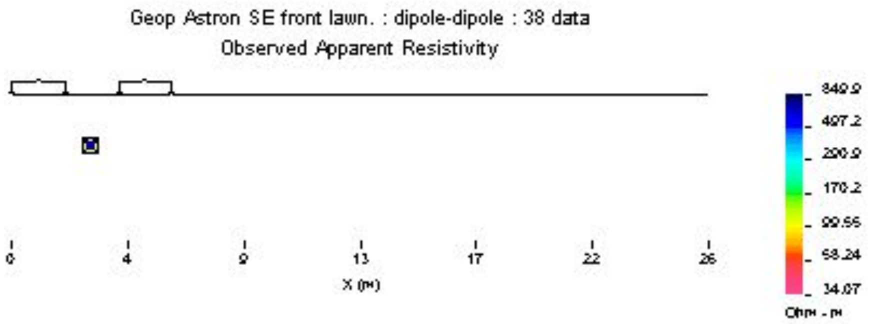
# How do we estimate the resistivity? → Apparent Resistivity




Does this values accurately represents the "true" distribution of subsurface resistivity?

$$\rho_a = k\frac{V}{I}$$

ρa: apparent resistivity (Ωm)

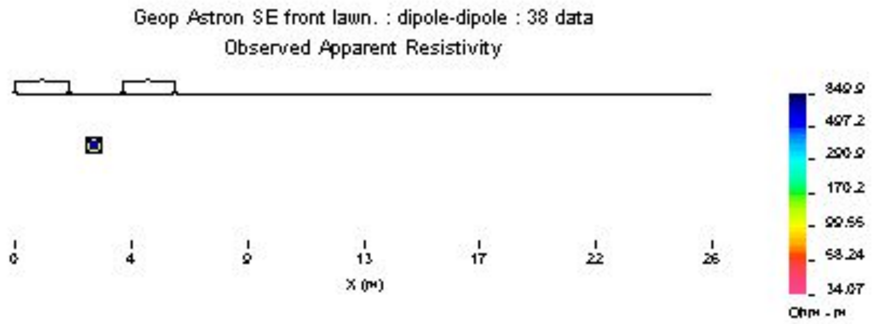
k: Geometric factor, represents the distance between the different electrodes

U: Potential (V)

I: Current (A)

It is based on the assumption that the ground is homogeneous and isotropic (homogeneous half space)

# How do we estimate the resistivity? → Apparent Resistivity




$$\rho_a = k\frac{V}{I}$$

ρa: apparent resistivity (Ωm)
k: Geometric factor, represents the distance between the different electrodes
U: Potential (V)
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It is based on the assumption that the ground is homogeneous and isotropic (homogeneous half space)

Does this values accurately represents the “true” distribution of subsurface resistivity?

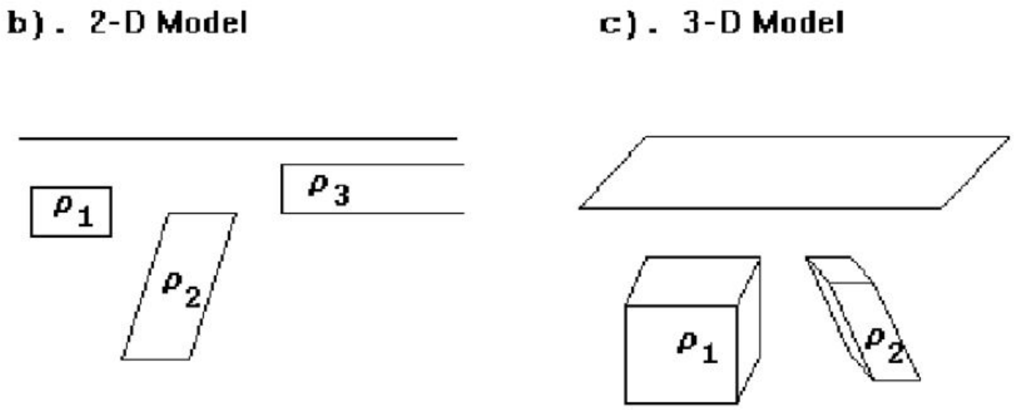
That apparent resistivity value would correspond to the true resistivity if the subsurface were homogeneous.

It is a fictitious value, it does not represent the actual resistivity of any specific layer

If the ground is layered or has heterogeneities, the apparent resistivity is just an integrated or weighted average response of the actual resistivity of the subsurface

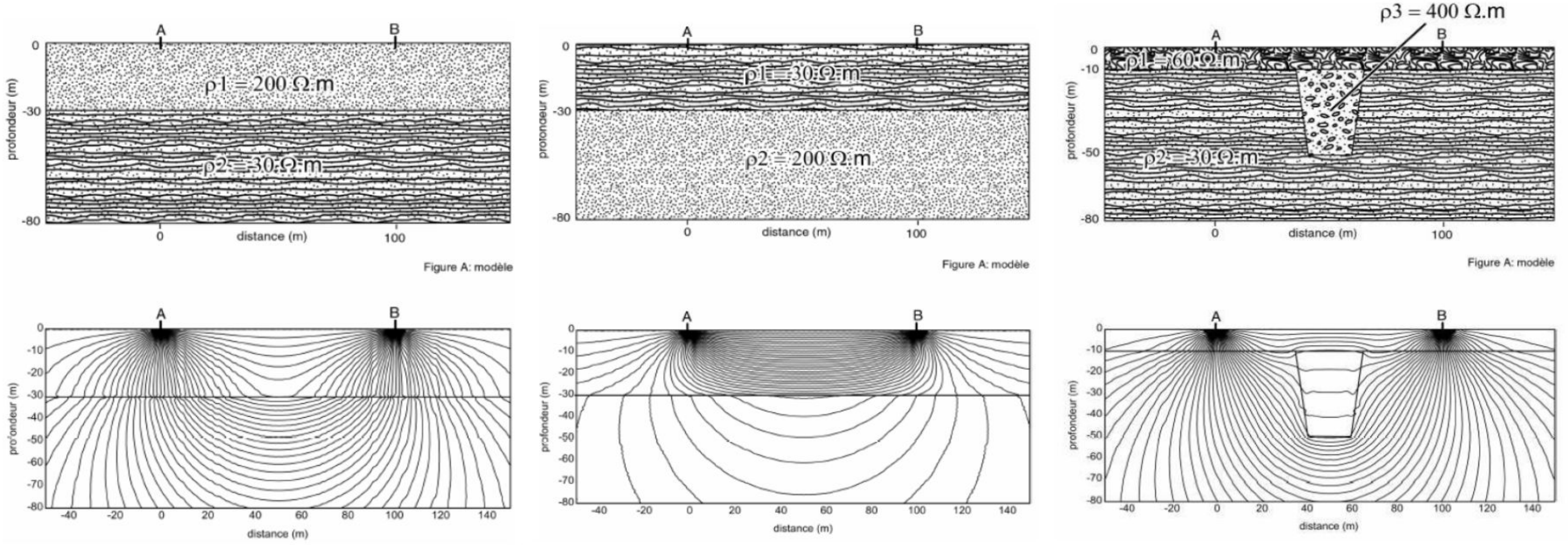
# Data Modelling

The subsurface is not an homogeneous half-space model.. it is more complex!



# Data Modelling

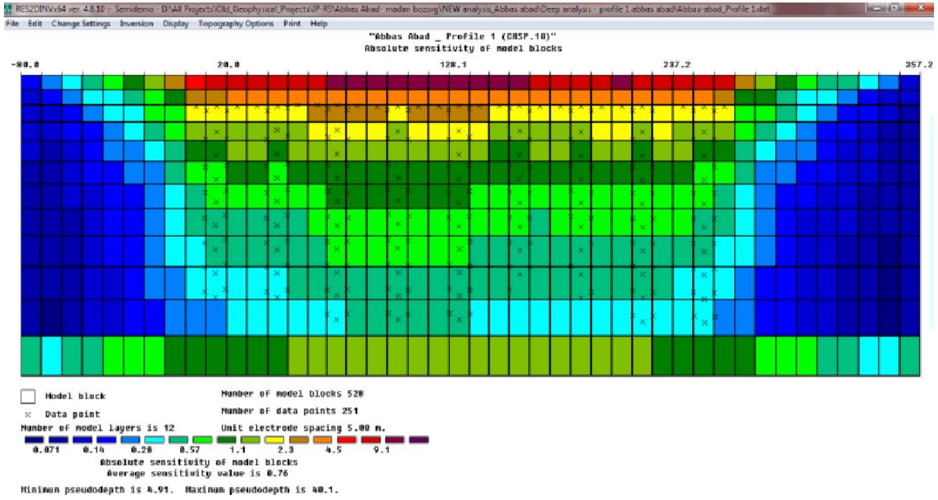
## Examples of 2D subsurface models



# ERT ⟶ Inversion procedure

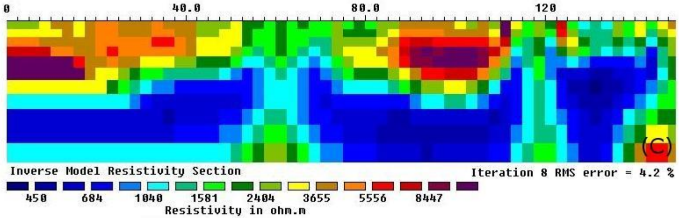
Inversion procedure⟶ estimate a physical parameter from field data

1. Subsurface is divided in a number of rectangular blocks
2. We assign a resistivity value to each block



# ERT ⟶ Inversion procedure

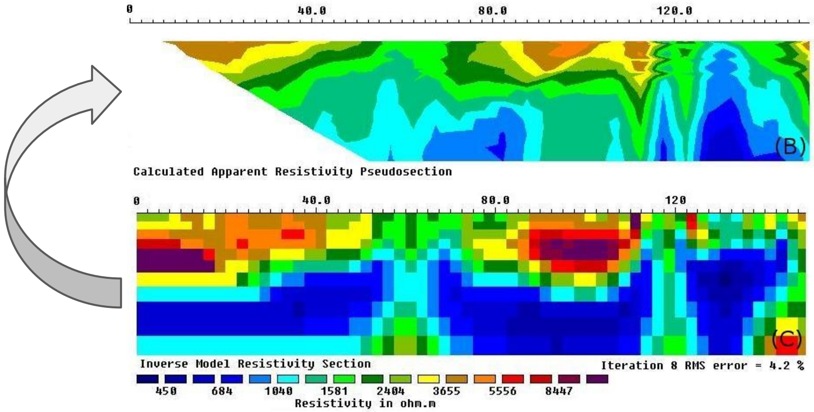
True resistivity values (model blocks)



# ERT ⟶ Inversion procedure

Calculated apparent resistivity values

True resistivity values (model blocks)



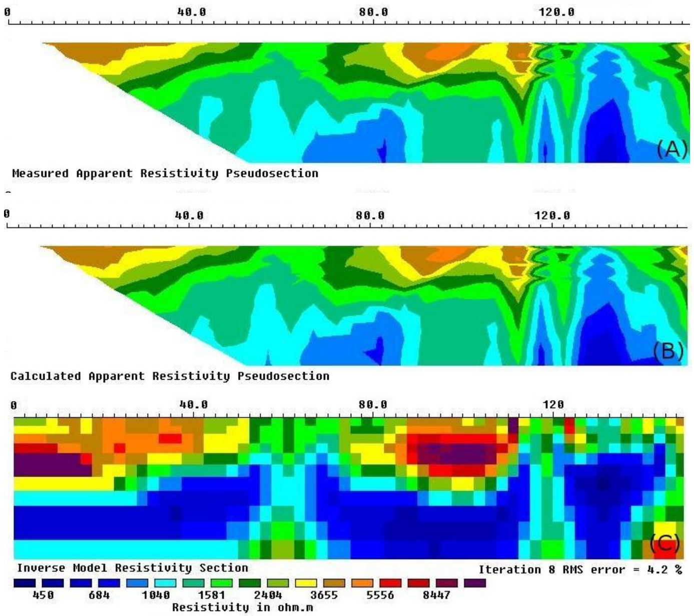
# ERT ⟶ Inversion procedure

Measured apparent resistivity values

?

Calculated apparent resistivity values

True resistivity values (model blocks)

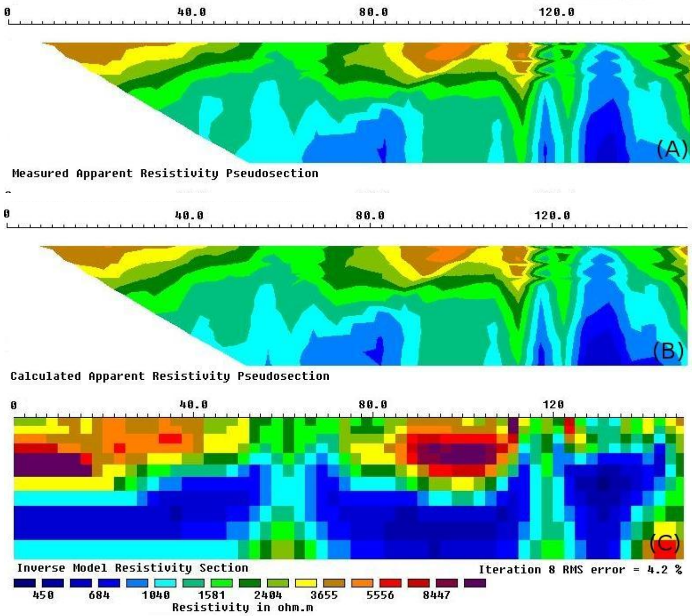


# ERT ⟶ Inversion procedure

Measured apparent resistivity values

Calculated apparent resistivity values

True resistivity values (model blocks)

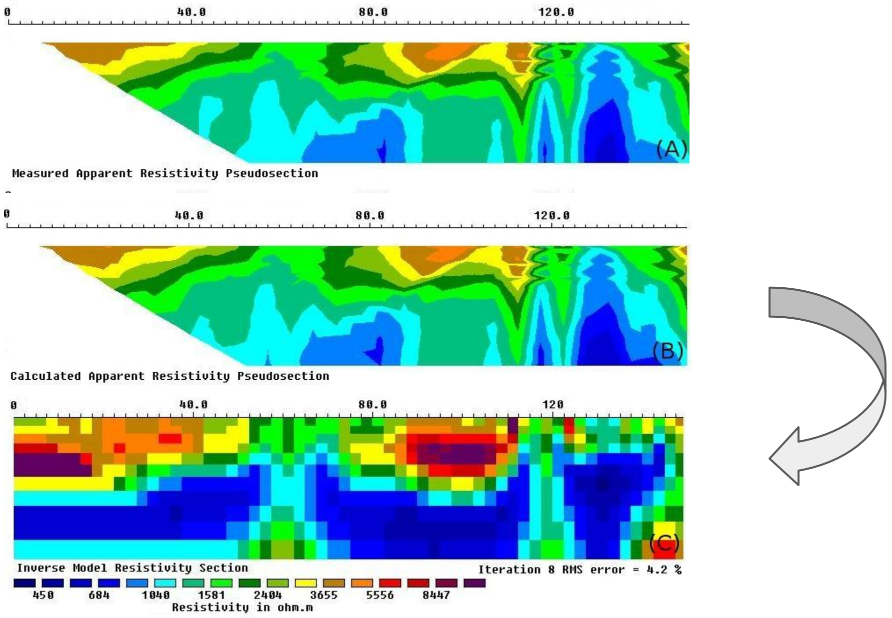


# ERT → Inversion procedure

Measured apparent resistivity values

Calculated apparent resistivity values

True resistivity values (model blocks)



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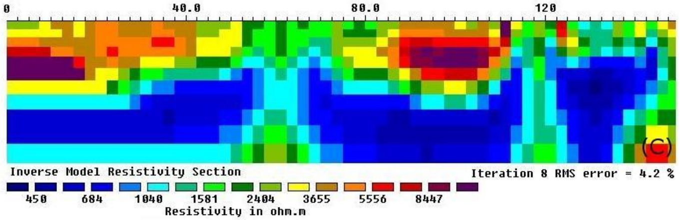
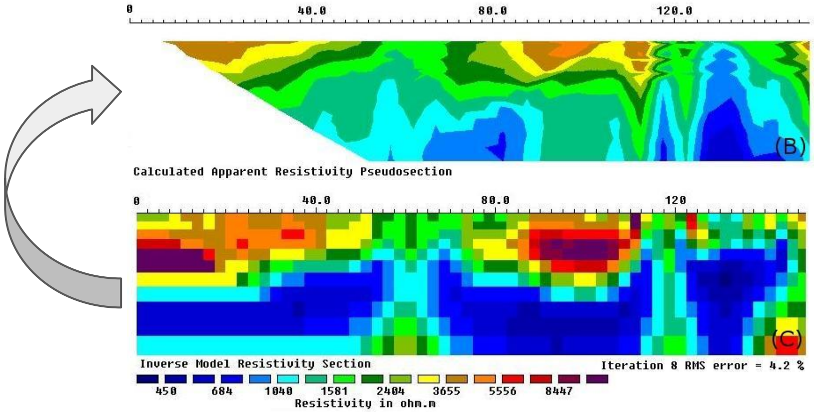
True resistivity values (model blocks)



# ERT ⟶ Inversion procedure

Calculated apparent resistivity values

True resistivity values (model blocks)



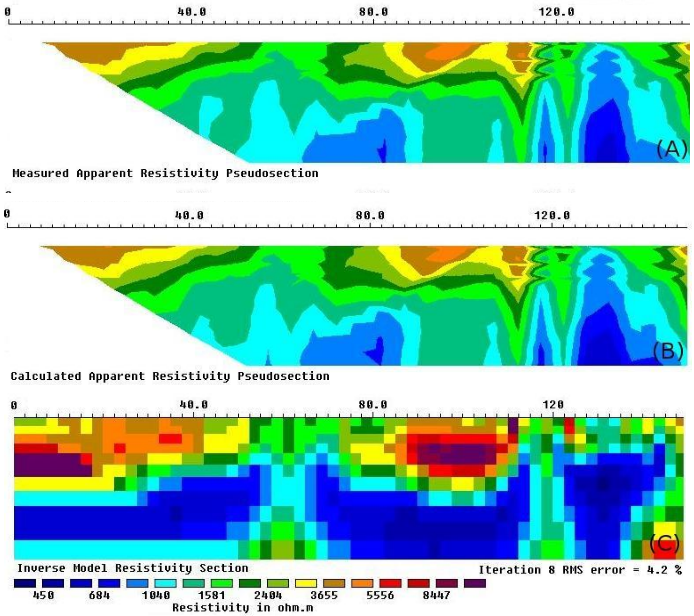
# ERT → Inversion procedure

Measured apparent resistivity values

= ?

Calculated apparent resistivity values

True resistivity values (model blocks)

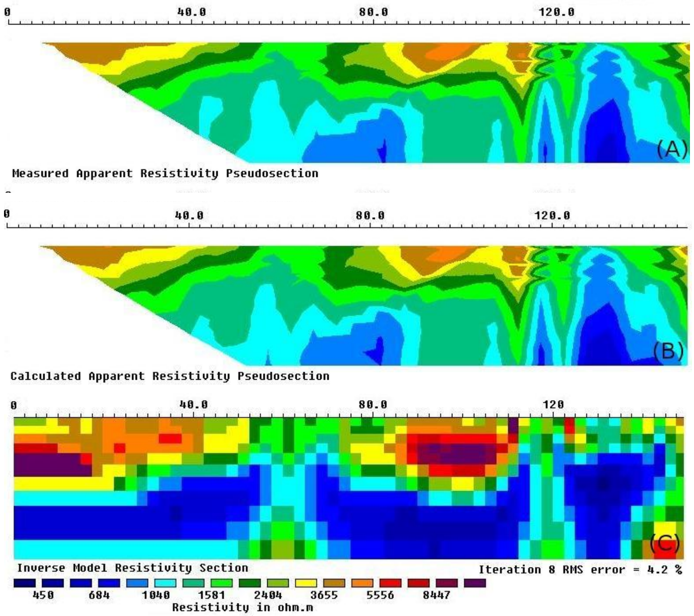


# ERT ⟶ Inversion procedure

Measured apparent resistivity values

Calculated apparent resistivity values

True resistivity values (model blocks)



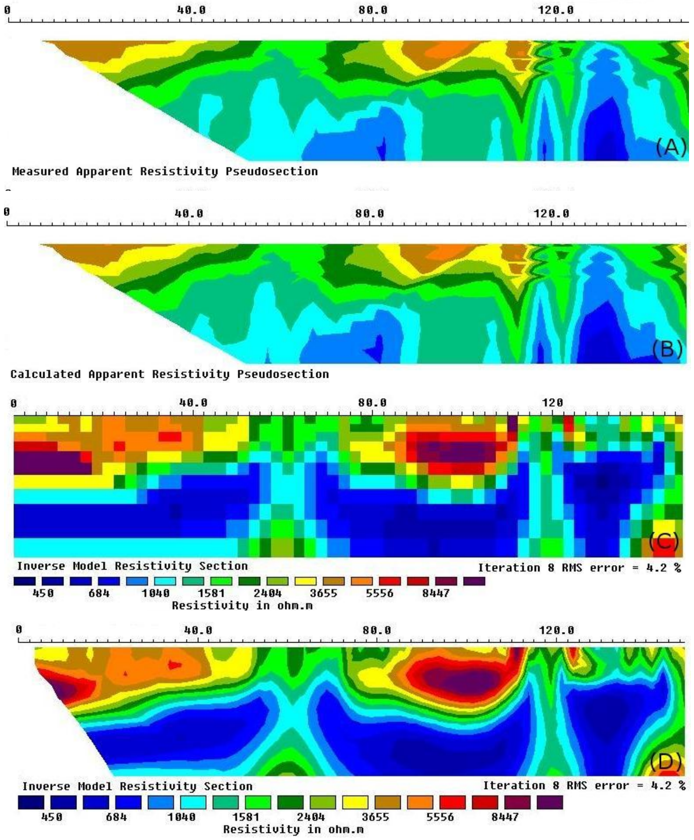
# ERT → Inversion procedure

Measured apparent resistivity values

Calculated apparent resistivity values

True resistivity values (model blocks)

True resistivity values (contoured section)



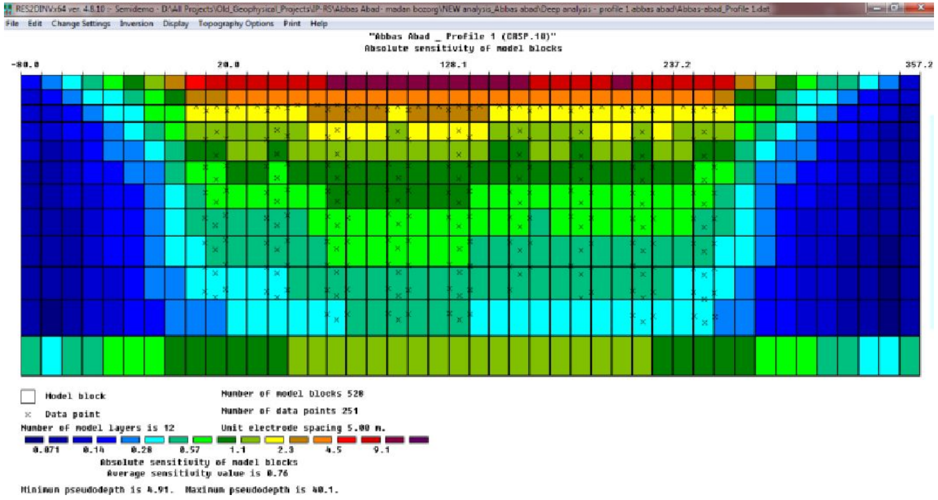
# ERT ⟶ Inversion procedure

## Inversion procedure⟶ estimate a physical parameter from field data

1. Subsurface is divided in a number of rectangular blocks

2. We try to determine the resistivity of the rectangular blocks that will produce an apparent resistivity pseudosection that agrees with the actual measured one

3. Optimization method tries to reduce the difference between the calculated and the measured apparent resistivity values by adjusting the resistivity of the model blocks




$$\min \sum_{i=1}^{N} \left(\rho_a^{\text{measured}}(i) - \rho_a^{\text{calculated}}(i)\right)^2$$

# Inversion approaches in RES2DINV

- We use RES2DINV to process resistivity data and build a 2D subsurface model.
  It works by inverting apparent resistivity measurements to estimate the true resistivity distribution underground.
- To do this, it adjusts a model to minimize the difference between measured and calculated data
- We can do this if different ways ( depending on the mathematical approach used in the inversion)

Choosing the right method depends on:

- The geological context
- The type of features we are trying to image
- The desired balance between resolution and noise sensitivity

# Inversion approaches in RES2DINV $\longrightarrow$ Two Common Minimization Methods:

- **1. Least-Squares Inversion (L2-norm)**

Minimizes the squared error between observed and calculated values:

$$\min \sum_{i=1}^{N} \left(\rho_a^{\text{measured}}(i) - \rho_a^{\text{calculated}}(i)\right)^2$$

- Produces smooth models (gradual resistivity transitions)
- It is best for continuous geology (sediment layers)

- **2. Robust Inversion (L1-norm)**

Minimizes the absolute error:

$$\min \sum_{i=1}^{N} \left|\rho_a^{\text{measured}}(i) - \rho_a^{\text{calculated}}(i)\right|$$

- Produces blocky models (sharper boundaries)
- It is best for targets with high resistivity contrasts (fractures, cavities, ore bodies)

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